How do we know leadership training works?

A bounded evidence map of evaluation criteria, measures and study designs · 20 October 2026

The central answer: specify what the training should change, measure that change with an appropriate instrument and rater, allow time for it to appear, and compare it with what would plausibly have happened without that training. “Works” can mean learning a skill, changing workplace behaviour, helping a team or improving a service. Those are different claims, requiring different evidence.

Two questions should stay separate: **What changed?** and **How confidently can we attribute the change to training?** A randomized study can measure only participants’ perceptions; an objective organizational outcome can still be confounded. Neither the outcome’s importance nor the study-design label answers both questions.

What researchers actually measure

A common organizing framework separates reactions, learning, workplace transfer and results. It is useful vocabulary, but it does not make these outcomes equivalent, guarantee a causal sequence or prescribe one universal success score. Programme goals may also concern leader identity, shared identity, relationships or readiness to lead.

Concrete aspects include communicating priorities, sharing and interpreting information, assigning roles, seeking team input, giving useful feedback, creating shared goals, and responding to implementation barriers. These examples come from the trauma-leadership rubric, principal-development evaluation and implementation-leadership evaluation. Their relevance depends on the role; they are not a universal list that every leader must score against.

Lacerenza and colleagues’ meta-analysis used the four categories across 335 adult-employee training studies. Its underlying designs included uncontrolled repeated measures as well as controlled comparisons; it excluded MBA and undergraduate samples and searched through December 2014. Its average effects support the possibility of benefit across varied programmes, not a guaranteed return for any particular course. The authors still warn about unavailable ineffective programmes despite tests that detected no publication-bias signal.

Question / criterion	Concrete measures and raters	Timing and interpretation
Was the programme usable, relevant and feasible? Participant experience and implementation	Relevance/acceptability ratings, attendance, completion, task submission, burden, interviews *Who:* Participants and programme records	During delivery and at completion. *Supports:* How participants experienced delivery and whether the planned programme was delivered or completed. *Does not establish alone:* Learning, changed behaviour, improved follower outcomes.
What did participants learn or become able to do? Knowledge, judgment and skills	Knowledge tests, scored cases, observed simulations; keep confidence/self-rated knowledge separate *Who:* Tests or trained assessors, with participant self-reports identified as such	Before training, after training, and retention assessment where relevant. *Supports:* Change on the particular task or measure. *Does not establish alone:* Transfer to an actual workplace or durable impact.

Question / criterion	Concrete measures and raters	Timing and interpretation
Did leadership behaviour change at work? Transfer and enacted leadership	Behaviourally specific employee/peer/supervisor ratings, structured observations, work samples or logs *Who:* People exposed to the behaviour and independent observers where feasible	A genuine pretraining baseline and follow-up after opportunities to apply the skills. *Supports:* Observed or perceived behaviour at stated times and contexts. *Does not establish alone:* Training caused the change; a generic 360-degree rating is not automatically a validated or unbiased measure.
Did followers or teams benefit? Relationships, shared identity, team functioning and well-being	Follower-reported support, team psychological safety, identification, coordination, stress or engagement *Who:* Team members where the claim concerns team experience; leader-only ratings remain leader perceptions	Baseline and repeated follow-up with stable or documented team membership. *Supports:* The measured experiences or functioning of the sampled followers/teams. *Does not establish alone:* Objective performance, universal benefit or causation.
Did the programme improve outcomes that matter in this setting? Service, educational, organizational or clinical outcomes	Student achievement, retention, service quality, errors, patient-care processes, costs *Who:* Administrative records, independent observation or explicitly identified subjective reports	Enough follow-up for the proposed mechanism, with relevant baseline or comparison data. *Supports:* Changes or differences in the particular outcome. *Does not establish alone:* Attribution to training: objective data can still be confounded.
Did change last, and were there costs or unintended consequences? Retention, maintenance, burden and harms	Repeated outcome measurements, staff turnover, workload, adverse experiences, implementation costs *Who:* Participants, followers and records, chosen for the specific consequence	Prespecified follow-up and ongoing adverse-event/burden collection. *Supports:* Durability and the observed tradeoffs within the follow-up window. *Does not establish alone:* No harm when harms were not assessed, or return on investment without both costs and attributable benefits.

This table is a lead synthesis of the selected evidence, not a validated instrument. Select the rows that match the programme's intended change; measuring everything is neither necessary nor automatically better.

Concrete instruments-and what their scores mean

A scale name is only the beginning. Check its operational definition, item wording, scoring, rater, context and evidence that it measures the intended construct. Internal consistency, agreement between raters and construct validity answer different questions. An adaptation from observed behaviour to "I know how" also changes what is being measured.

Instrument or approach	Actual use in an evaluation	Important limitation
Six programme-specific knowledge items; 13-item iLead behaviour scale	Managers rated perceived knowledge; employees rated implementation-leadership behaviour. The behaviour scale was administered only at follow-ups. Source	A study called pre/post may lack a baseline for a specific outcome. Self-rated knowledge is not an examination score.
Adapted Identity Leadership Inventory (ILI) and team-functioning scales	The 5R trial used three adapted items about knowing how to lead, plus leader-rated team engagement and teamfulness. Source	Team-labelled outcomes were leaders' perceptions. No team-member ratings were collected in this trial.

Instrument or approach	Actual use in an evaluation	Important limitation
69-item role-specific multisource feedback (MSF)	Educational medical leaders, heads, colleagues and trainees rated technical skills, human skills, citizenship behaviour and administrative skills. Source	These are role-specific domains. Rater selection, high baseline scores and attrition matter; both groups received feedback.
Behaviour rubric scored from real trauma-resuscitation videos	Separate blinded coders assessed enacted leadership and care processes; leadership rubric maximum 38. Source	Context-specific observed behaviour is stronger transfer evidence than confidence ratings, but it is not a universal leadership scale or proof of mortality benefit.
MLQ, evidence-based-practice, self-efficacy and self-esteem questionnaires	A nurse-leader feasibility trial used several established scales as participant self-reports. Source	Familiar instrument names do not turn a small feasibility trial into definitive evidence of patient or follower benefit.
Brief Job Stress Questionnaire	A workplace trial measured workers' job stressors, support and psychological distress. Source	Follow the outcome's own comparison and analysis unit. An outcome can be important while showing no detected programme benefit.
Administrative school outcomes	A principal-development trial used student test and staff-retention records alongside surveys. Source	An administrative outcome has its own denominator and follow-up coverage; it does not directly measure feedback quality.
16-item Leader Identity Measure	Four dimensions: strength of leader self-concept, integration across roles, meaning of leadership and inclusiveness/group orientation. Source	Scale-development evidence is not evidence of training responsiveness, workplace competence or a causal training effect.
Reflective assessment of leadership identity	Explore how learners understand leadership and themselves within individual, relational and collective processes. Source	Reflection can serve development and meaning-making. It is not a completed validated stage scale or behavioural performance test.

The two full-text measurement foundations concern identity assessment. One discusses student-leadership development and reflective evaluation; the other develops a measure across student and working-adult samples. Neither establishes responsiveness to training or a universal success cut-off. The original ILI and Clinical Teamwork Scale validation papers remained access-limited and were not fully appraised. Adapted ILI use above was examined directly in the 5R trial.

What study designs can establish

Evidence pattern	Useful conclusion	Remaining question
Post-course satisfaction or perceived-learning survey	Participants' experience and self-appraisal at that time	Did competence or behaviour change, and compared with what?
Before/after measures in one group	Change occurred on the measured outcome among those observed	Could maturation, other changes, expectations or selective dropout explain it?
Comparable nonrandom groups, baseline and follow-up	A more informative comparative change estimate	Were the groups already different in motivation, support or opportunity?
Random allocation with appropriate analysis and follow-up	Stronger attribution to the assigned programme relative to that comparator	Were outcomes valid, missing data consequential, clustering handled, and follow-up sufficient? Does the result generalize?

A waitlist compares training with waiting; an active comparison asks whether the programme adds value beyond another activity. Allocation of schools, workplaces or cohorts must be distinguished from measurement of individuals or events. The same trial may provide controlled evidence for one outcome and only uncontrolled evidence for another.

Examples that show why these distinctions matter

Positive experience can coexist with uncertain transfer

The iLead evaluation found favourable reactions and rising perceived knowledge, but no expected advantage for a contextualized training version. Both groups were trained and were assigned nonrandomly. Employee behaviour ratings lacked a pretraining baseline. Interviews suggested that reorganization, readiness and support affected implementation; these explanations were not experimentally established moderators.

A randomized trial may still measure only perceptions

The 5R identity-leadership trial randomized 12 cohorts, enrolling 76 leaders; 43 provided end-of-programme data. It reported improvements in perceived identity-leadership knowledge and some team-functioning perceptions relative to a waitlist. All came from the participating leaders. The study did not directly establish follower experiences, objective team performance or lasting effects. Cohort allocation, individual-level analysis, attrition and internally inconsistent reported statistics limit precision; no precise effect-size claim is made here. Its theoretical emphasis was shared “we/us” identity; a stronger individual “I am the leader” identity was not the programme’s universal goal.

Observed workplace behaviour and service outcomes can diverge

The trauma-team leadership trial used blinded video assessment of actual resuscitations after simulation training. It reported improved leadership behaviour, while the patient-care process comparison did not establish a clear benefit; the study was not powered for that care outcome. Follow-up events were generally close to training, so durability remains open. The paper contains conflicting effect-size and p-value reporting between its abstract and main text/table. The ledger preserves both; this report does not select the more favourable number or infer a mortality benefit.

More training does not guarantee better organizational outcomes

The IES principal-development evaluation randomized 100 schools within matched pairs. Principals received nearly 200 hours over two years, while comparison principals retained usual district development opportunities. The report found no average student-achievement benefit and some adverse teacher-reported feedback/support outcomes. It used administrative outcomes alongside surveys and time logs; feedback quality was not directly observed. The 100 schools describe the study, not every outcome’s analytic denominator; follow-up coverage varied. This is a programme-specific result in participating schools, not a verdict on all leadership education.

A training label can conceal different comparisons for different outcomes

The web-based supervisor-training trial randomized workplaces and measured worker stress/support outcomes in both arms, with no detected benefit on the main scales. Supervisors’ perceived knowledge was assessed only in the training arm. Its improvement therefore does not inherit the controlled design of the worker-outcome comparison. The small number of workplaces and analysis of clustered observations also limit precision and inference.

Multisource feedback is informative, but not a guarantee

A controlled medical-education leadership study found high course ratings without a demonstrated added course effect on one-year multisource leadership ratings. Both arms received feedback; allocation was nonrandom and follow-up incomplete. A separate hospital 360-degree intervention had no external comparator and found no significant change in its small supervisor subgroup. Its ratings involved feedback/reconciliation meetings, so multiple sources did not ensure independent judgments. These examples support measuring perspectives and transfer carefully; a nonsignificant result does not prove exact equivalence.

What the broader reviews add

A healthcare umbrella review included 86 reviews. Behavioural outcomes appeared in 36 and organizational/clinical outcomes in 30. Those are review-level reporting counts, not independent trials, success rates or percentages of programmes that work. The review found heavy reliance on self-report and limited long-term/system-level evidence. Its own extraction and risk-of-bias methods had limitations, including single-reviewer extraction and an AI-assisted appraisal step.

The medical-leadership systematic review distinguished self-reported from observed outcomes and found limited controlled evidence. The frontline clinical-leadership review likewise found that evaluations often omitted important outcome and design details. These reviews overlap: the umbrella cites both, and the medical review builds on earlier review data. We do not add their study totals or treat their conclusions as independent replications.

Promising programme features-practice, feedback, coaching, spacing or organizational support-can be associated with better results without being proven causal ingredients. The selected evidence supports checking whether learning opportunities and workplace opportunities match; it does not establish that one delivery recipe works everywhere. Costs, harms and long-term return on investment remain incompletely assessed.

A practical evaluation plan

The following is a planning proposal derived from this evidence, not an empirically validated checklist:

- **State the intended change.** Define the leadership problem, whose behaviour should change, and who should benefit. Choose a small number of primary outcomes before looking at results.

- **Match each goal to a measure.** Record the exact construct, instrument or rubric, scoring, rater and evidence of validity in a comparable context. Separate confidence from demonstrated competence.

- **Get the right baseline and follow-up.** Check each outcome individually. Assess immediate learning when appropriate, then allow opportunities for workplace use and reassess maintenance. There is no single universally correct follow-up interval.

- **Choose a credible comparison.** Use randomized or carefully justified comparison groups where feasible; distinguish usual practice, waiting and active alternatives. Track dose, fidelity, competing changes, missingness and clustered assignment.

- **Report the whole result.** Include effect estimates and uncertainty, practical importance, null and adverse findings, attrition, costs and limits. Avoid selecting one positive subscale among many as proof of overall success.

For example, a course teaching constructive feedback could assess a scored practice conversation immediately, obtain employee reports or independent observations after workplace opportunities, and compare change with a suitable alternative. Retention or productivity would be additional claims

requiring their own time horizon and attribution evidence. This is an illustrative design proposal, not an evaluation already run.

Scope, source coverage and review

Discovery used six bounded OpenAlex query families, with 20 relevance-ranked hits each: 120 raw hits and 119 unique provider IDs. These are discovery candidates, not a comprehensively screened field census. Targeted specialist searches and source lookups added a small purposive set to cover contrasting designs, settings and measurements. The previous 951 adaptive-leadership records were not the sampling frame and were not rescanned.

The final appraisal set contains 14 distinct sources: 4 reviews, 8 primary evaluations and 2 measurement/theory foundations. Actual accessible full-text sections were examined; read scope and hashes are recorded per source. Three further candidates-the Dvir field experiment, original ILI validation and Clinical Teamwork Scale validation-remained access-limited and were excluded from detailed appraisal. Source selection is purposive, English-readable and availability-dependent, with substantial healthcare representation. No new pooled effect, prevalence estimate or ranking of programmes was calculated.

Protocol versions document the original plan, clarification of metadata versus source lookups, expansion of the targeted-source ceiling from eight to twelve, and explicit inclusion of authoritative government primary evaluations. Targeted web discovery exceeded the additional-source ceiling; this deviation is retained in the search audit rather than retroactively redefining the cap. Full-text appraisal stayed below the maximum of 24 sources. Web discovery was not exhaustively enumerated, and no complete web-result count is claimed. Search receipts and unsuccessful acquisitions remain in the audit trail. Possible primary-study overlap across reviews is explicitly flagged; a complete review-by-primary-study crosswalk was not reconstructed.

Work used native Codex orchestration: a scientific lead, separate efficacy/design and measurement specialists, then separate methods and statistical/reproducibility review. Specialists contributed research; they were not represented as independent final reviewers. This was not a controller-managed MCP research session. Human domain-expert assessment: not assessed. Candidate 2 received separate methods and statistical/reproducibility review and was accepted within the stated scope limits, with no unresolved major objections. This final report changes only the administrative review-status statement; human domain-expert assessment remains not assessed.